



Do ADHD Symptom Clusters Moderate the Association Between Autistic Traits & Error-Related Negativity in Children With and Without Autism?: An EEG/ERP Study

Elia F. Soto^{1,2} · Marie Shuter² · Nicole Shea² · Ashley Schiros² · Jayden Nord¹ · Yvonne C. Kuo² · Huilin Sun² · Kevin Antshel² · Natalie Russo²

Received: 13 February 2025 / Accepted: 26 April 2026
© The Author(s) 2026

Abstract

Purpose This study aims to address existing mixed findings in the event-related potential error-monitoring (i.e., error-related negativity; ERN) literature in children with autism spectrum disorders (autism). This study is the first to examine the moderating roles of parent-reported attention-deficit/hyperactivity disorder (ADHD) symptoms (i.e., inattention/hyperactivity/impulsivity) on the relation between autistic traits and the ERN/ Δ ERN in children with and without autism.

Methods Data was collected from 40 children (10–17 years of age) diagnosed with ($n=17$) and without ($n=23$) autism matched on age and IQ. Participants completed a flanker task while ERPs were measured. Autistic traits were measured on the Autism Quotient, while ADHD symptoms of inattention and hyperactivity-impulsivity were assessed via parent report on the BASC-2/3 Attention Problems and Hyperactivity subscales. Four moderation models were conducted examining the separate effects of each ADHD symptom cluster on autistic traits predicting the ERN and Δ ERN mean amplitudes.

Results Results indicate that the autism group had higher reported Attention Problems and Hyperactivity scores, lower accuracy scores on a traditional Flanker task, and similar reaction times (RT) compared to neurotypically developing peers. Attention Problems moderated the relations between autistic traits and both the ERN and Δ ERN, while Hyperactivity only moderated the relation between autistic traits and the ERN.

Conclusions This study provides an important first step towards understanding how co-occurrence between autism and ADHD symptoms impact neurocognitive processes, such as error-monitoring. These findings suggest future works need to control for co-occurring ADHD symptomology when investigating ERPs in autism samples.

Keywords Autism · Autism spectrum disorder · ADHD · ERN · Attention · Hyperactivity

The clinical utility of event-related potentials (ERPs), a temporally sensitive measure of brain activity, to identify risk and predict developmental trajectories of psychopathology has garnered increasing attention by researchers (Hajcak et al., 2019; Perkins et al., 2022) and clinicians (Jadhav et al., 2022). One commonly examined ERP associated with

error-monitoring, the error-related negativity (ERN), has been repeatedly examined across two of the most prevalent neurodevelopmental disorders: Autism Spectrum Disorder (autism) and Attention-Deficit/Hyperactivity Disorder (ADHD; CDC, 2023). Autism has high co-occurrence with ADHD (50–70%; Rong et al., 2021; Grzadzinski et al., 2016) at diagnostic and trait/symptom levels. Both disorders are associated with executive functioning difficulties (see Kofler et al., 2024), which the ERN is purported to measure (Downes et al., 2017). Despite high overlaps in autism and ADHD symptomology, the ERN has largely been examined in each of these neurodevelopmental disorders in isolation. Thus, little is known about how the co-occurrence of these conditions impact the ERN during childhood. An important step towards filling these gaps in the literature is to examine

✉ Natalie Russo
nrusso@syr.edu

¹ Department of Psychology, Louisiana State University, Baton Rouge, USA

² Department of Psychology, Marley Education Center, Syracuse University, 765 Irving Ave, Syracuse, NY 13210, USA

how the relations between autism and the ERN is impacted by ADHD symptomology. The present study is the first to examine the effects of co-occurring inattention and hyperactive-impulsivity symptom clusters on the relation between autism and the ERN.

Utility of the ERN

The ERN is a specific response-locked ERP component observed in the frontal central scalp region at approximately 50–100 ms after making an error (Falkenstein et al., 1991; Gehring et al., 2012). Another closely related ERP to the ERN is the Δ ERN, which is derived from the difference of activity voltages in response to incorrect and correct trials (error – correct; Gehring et al., 1993). Both are theorized to reflect unconscious error processing/monitoring and have been moderately to strongly associated with behavioral and rating-scale indicators of post-error-processing (Cavanagh & Shackmann, 2015; LoTemplio et al., 2023). The Δ ERN is theorized to represent the unique and purer neural activity attributed to error monitoring (Simons, 2010) and demonstrates better convergent validity when examined across several different response-based tasks (i.e., Flanker, Go/Nogo) compared to the ERN (Riesel et al., 2013).

Neuroanatomically, the ERN/ Δ ERN are hypothesized to generate in the anterior cingulate cortex (ACC; Mathewson et al., 2005), a brain area associated with conflict and outcome monitoring (Botvinick et al., 2004). The ERN/ Δ ERN have been reliably observed in young children (as early as four-years-old) into late adulthood, with the ERN/ Δ ERN becoming larger and more negative over time (Davies et al., 2004; Downes et al., 2017). One study demonstrated strong psychometric properties (including high test–retest reliability, convergent validity, and internal consistency reliability) of the ERN in children between the ages of 8–13 years (Meyers et al., 2014). Both the ERN and Δ ERN are strongly linked with behaviors associated with executive dysfunction. More specifically, larger and more negative ERNs/ Δ ERNs in healthy individuals have been linked with greater behavioral inhibition, greater avoidance behaviors, and reduced impulsivity (McDermotte et al., 2009; Meyer et al., 2012; Potts et al., 2006). Taking the measurement strengths, developmental sensitivities, and moderate-to-strong associations with behavioral outcomes, the ERN and Δ ERN may serve as promising bioindicators of mental health across autism and ADHD. While existing literature has examined the ERN/ Δ ERN in each condition separately, very little is known presently about how their co-occurrence may impact the ERN.

Autism and the ERN/ Δ ERN

Autism is characterized by social communication difficulties and restricted repetitive behaviors and/or interests (APA, 2022). Early autism models postulate that autistic individuals experience difficulties with self-monitoring (Russell, 1997), which contribute to social-emotional and social-cognitive challenges (Henderson et al., 2006). Further, theoretical frameworks, such as the executive dysfunction theory of autism postulate that difficulties with perseveration and rigidity may stem from executive functioning challenges with (1) recognizing when to and successfully disengage from a task and (2) initiating a new action or non-routine. These frameworks and models garnered some support from child and adult studies demonstrating that autistic individuals evidence reduced error-monitoring and increased behavioral difficulties after making an error (Bogte et al., 2007; Thakkar et al., 2008). However, relatively few studies have examined the ERN/ Δ ERN in autistic child populations (Groen et al., 2008; Henderson et al., 2006). Findings have been largely mixed, showing reduced (Sokhadze et al., 2010), increased (Henderson et al., 2006), and no ERN/ Δ ERN mean amplitude differences (Groen et al., 2008) in autistic individuals compared to neurotypically developing (NTD) peers. These inconsistent findings are further complicated by several notable methodological shortcomings, such as lacking valid, reliable, gold-standard assessments for determining a diagnosis of autism (i.e., ADOS-2; Lord et al., 2012), failing to account for high co-occurrence of co-occurring symptomology (i.e., inattention, hyperactivity-impulsivity), and not taking into account impacts of IQ and/or age (see Burack et al., 2004 for review). These concerns in current literature base on ERN/ Δ ERN and autism substantially limit our understanding of error-monitoring in autism.

ADHD and the ERN/ Δ ERN

ADHD is characterized by symptoms of inattention and hyperactivity-impulsivity and is the most prevalent neurodevelopmental disorder (APA, 2022). Similar to autism, Barkley's (2011) theory of executive dysfunction posits that ADHD symptoms stem from difficulties of behavioral inhibition, which ultimately impact executive functioning and self-regulation—both of which are linked with the ERN/ Δ ERN. Results examining the ERN/ Δ ERN within child ADHD populations have been largely mixed (Meyer & Hajcak, 2019; Shiels & Hawk, 2010). One study shows that children with ADHD have larger ERNs/ Δ ERNs compared to NTD peers (Burgio-Murphy et al., 2007). One small meta-analysis shows that adolescents and adults with ADHD

have smaller ERNs/ Δ ERNs compared to NTD individuals (Geburek et al., 2013) while another recent meta-analysis suggests there are no discernable ERN/ Δ ERN differences in individuals with ADHD and NTD peers after controlling for age (Kaiser et al., 2020). Examining the ERN/ Δ ERN and ADHD using more dimensional approaches has yielded similar mixed results, such that more ADHD symptoms have been related to smaller ERN/ Δ ERN amplitudes in some adult samples (Marquardt et al., 2018;) while no differences have been observed in other adult samples and in some child samples (Herrmann et al., 2009; Wiersma et al., 2009).

Co-Occurring Autism and ADHD and the ERN/ Δ ERN

In addition to their high co-occurrence rates, Autism and ADHD have both been linked with executive dysfunction, albeit in different ways (i.e., perseveration vs impulsive behaviors), that ultimately impact self-monitoring (Barkley, 2011; Hill, 2004). Within ERN research across both autism and ADHD, the majority of studies have been conducted in adults and include notable methodological shortcomings (i.e., examining wide age ranges, not assessing for or controlling for co-occurring conditions or symptoms). Taken together, much of the existing literature-base on the ERN/ Δ ERN is largely convoluted, potentially due to the large co-occurrence between autism and ADHD. An important, systematic, first step towards clarifying the role of the ERN/ Δ ERN in these disorders is to examine the relation between the ERN/ Δ ERN in autistic youth compared to an IQ and age matched NTD control group and then examining how symptoms of co-occurring/comorbid conditions (i.e., inattention, hyperactivity-impulsivity) impact that relation.

Current Study

Existing literature examining error monitoring as assessed with the ERN and the Δ ERN in autism and ADHD is highly mixed, leaving notable gaps in our understanding of the ERN/ Δ ERN. The current study fills these gaps by being the first to examine the moderating effects of commonly co-occurring ADHD symptom of inattention and hyperactivity-impulsivity on the ERN/ Δ ERN within children diagnosed with autism and comparison children who are group-matched on both age and IQ. Further, the current study is the first to take both a dimensional and categorical approach to obtain a more comprehensive understanding of how the ERN/ Δ ERN, inattention, and hyperactive behaviors relates to a diagnosis of autism (categorical) and autistic

traits (dimensional). We hypothesize children with autism will have more autistic traits as well as have higher parent-reported ratings of both ADHD symptom clusters compared to children without autism. Additionally, given the prevalent mixed findings in the literature (Pudlo & Pisula, 2022; South et al., 2010), we aim to examine between group differences on performance-based measures on a traditional arrow Flanker task among children with and without autism. Using neuro-based measures, we hypothesize that children with autism, compared to their NTD peers, will have larger mean amplitudes on the ERN. This hypothesis is consistent with existing theoretical EF models, suggesting that autistic individuals show larger difficulties with perseveration and rigidity. Lastly, and consistent with Barkley's ADHD theoretical models of EF, we hypothesize the relation between autistic traits predicting the ERN will be weakened by both ADHD symptom clusters using four separate moderation models, such that the higher the symptom cluster of ADHD, the weaker the relation between autistic traits and the ERN and the Δ ERN.

Methods

Participants

The study sample was comprised of 40 children (16 girls), aged 10–17 years old ($M=14.19$, $SD=2.00$) from the Northeastern US. Participants were recruited through community resources from 2017–2024 for participation in a larger study of the neurological sensory processing mechanisms underlying autism and other related neurobiological disorders. All 40 children met inclusion criteria for this study; including being between the ages of 10–17 years old and either meeting criteria for the autism group or the NTG group as outlined below. Children were excluded for gross motor impairment, seizure disorder, psychosis, or intellectual disability/estimated IQ at or below a standard score of 80. Notably, data collection was temporarily halted during 2020–2022 due to COVID-19. Sample ethnicity/race included 36 White Non-Hispanic (90%), two multiracial (5%), one Black (2.5%), and one Asian children (2.5%) (Table 1). No participants reported identifying with a Hispanic ethnicity.

Group Assignment

Children were assigned into one of two groups: autism and NTD. Children suspected of or diagnosed with autism completed a comprehensive psychological evaluation to confirm criteria was met for the autism group. The autism group ($n=17$) met all of the following criteria: (1) *DSM-5* diagnosis

Table 1 Descriptive Statistics

Variable	Aut (n=17)		NTD (n=23)		<i>t</i>	η^2	<i>p</i>	Cohen's <i>d</i>
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>				
Sex (Male/Female)	(12/5)		(12/11)		–	1.38	.24, <i>n.s.</i>	–
Med Status (Yes/None)	(11/6)		(0/23)		–	20.53	<.001	–
Race (A/B/M/W)	(0/1/1/15)		(1/0/1/21)		–	2.15	.54, <i>n.s.</i>	–
Age (years)	14.30	2.24	14.11	1.85	0.30	–	.77, <i>n.s.</i>	–
PRI	112.29	12.40	110.96	16.07	0.29	–	.78, <i>n.s.</i>	–
Rating Scales								
AQ Total Score	33.47	4.93	12.13	5.47	12.71	–	<.001	4.07
BASC-2/3 Attention Problems	63.47	9.82	44.39	7.40	7.02	–	<.001	2.25
BASC-2/3 Hyperactivity	68.00	14.29	45.83	9.14	5.98	–	<.001	1.91
Performance Measures								
Flanker RT (msec)	0.73	0.34	0.73	0.24	–0.01	–	.99, <i>n.s.</i>	<0.01
Flanker Accuracy (%)	90.06	9.58	95.62	4.50	–2.45	–	.02	0.78
ERP Measures (mean amplitude)								
ERN at 0–100 ms	–1.50	3.54	–1.49	3.81	–0.01	–	.99, <i>n.s.</i>	<0.01
CRN at 0–100 ms	0.42	1.02	2.24	2.82	–0.73	–	.47, <i>n.s.</i>	0.23
ΔERN at 0–100 ms	–1.92	2.51	3.34	4.08	0.49	–	.63, <i>n.s.</i>	0.16

n.s.=non-significant; Aut=Autism; NTD=Neurotypically Developing; Med Status=Medication Status; Race: A=Asian/Asian American, B=Black/African American, M=Multiracial, W=White Non-Hispanic; PRI=WASI-Perceptual Reasoning Index; RT=reaction time

of autism by licensed psychologist, (2) elevations on a semi-structured clinical interview, ADI-R (Rutter et al., 2005) or Kiddie Schedule for Affective Disorders and Schizophrenia for School-Aged Children (K-SADS; Kaufman et al., 1997; 2013 Update), and (3) clinical elevations on the ADOS-2 administered or coded by research reliable clinicians (Lord et al., 2012). Medications were not withheld prior to testing, and medication status was recorded for all child participants. Only the autism group reported taking medications, which included antidepressants, antianxiety, antipsychotics, sleep aids, and stimulant medication. Co-occurring conditions in the autism group included ADHD, anxiety, depression, and oppositional defiant disorders. Co-occurring conditions are based on caregiver reports of their child's mental health histories.

The NTD group ($n=23$) consisted of children who did not meet autism criteria or any other psychiatric, neurological, genetic, or learning disorders as reported by caregivers on the KSADS and collected background history demographic information. Group characteristics of both the autism and control group are presented in Table 1. As expected, positive endorsement of medication status was higher in the autism group compared to the control group ($p<.001$). No other between group differences on demographic variables were observed (all p 's $>.05$).

Procedures

Testing occurred as part of a larger battery that involved 2–3 sessions of approximately 3 hours each. Questionnaires, ADOS-2 testing, and IQ testing were conducted during the

initial session while EEG testing was completed during the second session. Tasks were counterbalanced within the EEG testing session to minimize order effects. Children received breaks after each task to minimize fatigue. Performance was monitored by the examiner observing from an adjacent room out of the child's view to minimizing improvements related to examiner demand characteristics (Gomez & Sanson, 1994). Syracuse University's Institutional Review Board (IRB) approved the study, and all participating caregivers and children gave informed consent and assent respectively.

Measures

Intellectual Functioning (IQ)

All children were administered the Wechsler Abbreviated Scale of Intelligence-Second Edition (WASI-II; Hsiao-pin, 2011). Consistent with best-practice matching processes (Burack, et al., 2004) and prior autism child studies, the Perceptual Reasoning Index (PRI) was used to match participants on IQ (Cary et al., 2024; Matsuba et al., 2024).

Rating Scales

Autism Spectrum Quotient (AQ)

The AQ (Baron-Cohen et al., 2001) child, adolescent, and adult forms are each 50-item rating scales used to assess autistic traits. Across all three forms, item content is consistent and adapted for development levels, such that the parent-report Child AQ (Auyeung et al., 2008) ranges for

participants aged 4–11 years old, the parent-report Adolescent AQ (Baron-Cohen et al., 2006) for participants aged 12–15 years old, and the self-report Adult AQ for participants 16 years and older (Baron-Cohen et al., 2001). All three forms have cut-off scores that are highly suggestive of an autism diagnosis, are normally distributed, and measure autistic traits (Auyeung et al., 2008; Baron-Cohen et al., 2001, 2006). All items across all three forms can be scored using a binary scoring system, where items scored as 1 indicate positive autistic trait endorsement (Baron-Cohen et al., 2001, 2006). Total scores (range: 0–50) served as the primary indices of autistic traits for the current study, with higher scores reflecting a greater degree of autistic traits. All three forms have acceptable psychometric properties, including moderate-to-high internal consistency ($\alpha=0.63–0.97$), excellent test–retest reliability, and good convergent validity with autism diagnoses (Auyeung et al., 2008; Baron-Cohen et al., 2001, 2006).

Behavior Assessment System for Children (BASC-2/3)

The BASC-2/3 (Reynolds & Kamphaus, 2004, 2015) parent report forms are 150–175 items that assess internalizing and externalizing behavior problems in children ages 2–21. Raw scores are converted to age- and gender-specific T-scores based on the national standardization sample ($N=1,800$ per form). Parent reported Attention Problems and Hyperactivity subscale T-scores served as the primary indices of ADHD inattention and hyperactivity-impulsivity symptom clusters, respectively. Psychometric support for these subscales includes high internal consistency ($\alpha=.85-.95$), test–retest reliability ($r=.80-.92$), and concordance with other informant-based ratings of ADHD (Reynolds & Kamphaus, 2004, 2015; Sullivan & Riccio, 2006).

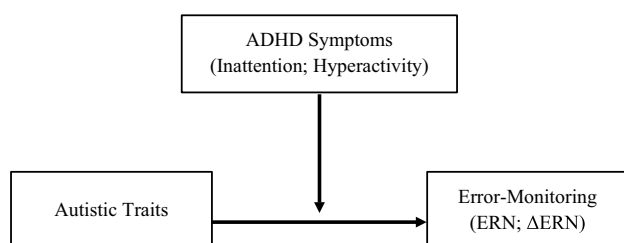


Fig. 1 Conceptual Moderation Model. Simple moderation model of autistic traits predicting error-monitoring (ERN or Δ ERN) as moderated by ADHD symptoms (Inattention or Hyperactivity)

Behavioral Performance Measure and Associated ERPs Indicators

Flanker

Participants completed a Flanker task that used sets of five horizontally aligned arrowheads (e.g., >>>>>). The task was programmed and initiated using Stream (Wyble, 2019), a MATLAB™ (MathWorks, Inc 2021) toolbox alongside Psychophysics toolbox (Brainard, 1997). Stimuli were presented on a VPixx monitor with a 22.5-inch display (resolution: 1920 × 1200 pixels; refresh-rate: 120 Hz). For each trial a fixation cross (+) was presented on the screen (200 ms), followed by arrowheads presented (50 ms), then followed by an inter-stimulus-interval (range: 600–1,600 ms; see Fig. 1). Arrowheads were presented in white Arial 30-point font with one space between each arrowhead against a grey background and were positioned in the center of the screen. Participants were instructed to press the “M” key on the keyboard when the center arrowhead was facing right and the “X” key on the keyboard when the center arrow was facing left. For 40% of the trials the center arrowhead was presented congruently (e.g., >>>>>) and for 60% of the trials the center arrowhead was presented incongruently (e.g., >><>>). There were four possible conditions: congruent right, congruent left, incongruent left and incongruent right (see Fig. 1). There were twelve experimental blocks, with each block containing 40 trials (480 total trials). RT and accuracy were recorded for all trials.

Electrophysiological Recording

Electrophysiological responses were continuously recorded using a Net Amps 300 amplifier utilizing Net Station Software and Electrical Geodesics, Inc. hardware. Participants were fitted with a high-density 128-channel elastic cap sensor-net. During online data collection, raw EEG data was sampled at 250/second and then digitized by a 24-bit A/D converter. Impedances were measured and kept below 50k Ω . Raw EEG data was processed offline and analyzed in MATLAB™ using EEGLAB (Delorme & Makeig, 2004) and ERPLAB (Lopez-Calderon & Luck, 2014).

ERP Processing and Extraction

Offline, data preprocessing steps were completed in the following order on a subject-by-subject basis: Independent component analysis (ICA), re-referencing, segmentation,

baseline correction, manual artifact rejection, and averaging. DC offsets were removed. Prior to the ICA, a bipolar HEOG and VEOG signals were derived from channels 31, 1, and 126, 4; respectively. Data were band-pass filtered using 0.1 Hz–30 Hz, 12 dB/oct roll-off, and non-causal Butterworth impulse response function. EEG segments containing large muscle artifacts, extreme voltage offsets, and/or break periods lasting >2 s were identified and removed by a semi-automatic ERPLAB algorithm. Visual inspection of waveforms and scalp distributions were conducted to identify and remove ICA components determined to be associated with eye movements (Jung, et al., 2000). Following ICA, data was re-referenced to the average of the mastoids (Luck, 2014). Data was segmented into epochs 200 ms pre-response and 400 ms post-response. Baseline correction was conducted using a window of 200 ms from -200 to 0 ms. Artifact detection involved ERPLab (Lopez-Calderon & Luck, 2014) to automatically remove epochs with activity greater than a threshold of 100 μ V (moving window: 200 ms, window-step: 100 ms). Visual inspection of remaining data was conducted to identify any outstanding noisy channels. Removed channels were interpolated using the surface spline interpolation methods (Perrin et al., 1987). Grand averages were calculated for each condition for each participant.

ERP data were extracted with ERPLAB (Lopez-Calderon & Luck, 2014) for each participant individually that made ≥ 6 errors and ≥ 4 accepted error trials that made it through the EEG processing pipeline. Averaged ERPs for each participant were visually reviewed by three of the authors blinded by participant group assignment for indiscernible averaged ERN waveform for each participant. In addition to the 40 participants included in the study, a total of six other participants were excluded from all analyses for having made <6 total errors upon task completion, nine for having <4 accepted error trials following EEG preprocessing procedures, and one for having an indiscernible averaged ERN waveform. Notably, there were no group differences on age, sex, PRI, medication status, or group membership between the 40 included and 16 excluded participants (all p 's $> .05$, Table S1). For all ERPs of this study, activity from electrode Fz was extracted. Consistent with ERN/ Δ ERN studies (Marsh et al., 2023; Olvet & Hajcak et al., 2009), the ERN was defined as the average voltage between 0–100 ms following the response on error trials. The correct related negatively (CRN; average voltage following a correct response) was extracted across the same time range to calculate the Δ ERN. Mean amplitudes were extracted for each ERP as mean amplitudes are less sensitive to noise compared to other measures of amplitude and reflect more than a single time point (Luck et al., 2005).

The Δ ERN was derived from the difference of the ERN and CRN mean amplitudes.

Data Analysis Plan

The current study investigated the extent to which a categorical diagnosis of autism is associated with performance and neurophysiological differences on a Flanker task compared to those without autism. Accordingly, our analytic plan was organized into two tiers. First, we conducted data screening/cleaning and examined the extent to which children with autism and without autism exhibit group differences on primary variables of interest (i.e., autistic traits, inattention, hyperactive behaviors, ERN, and Δ ERN) as hypothesized using t -tests.

Second, we conducted two separate moderation models using JASP0.17.1.0 (JASP Team, 2004) with 10,000 bias-corrected, bootstrapped samples per model to analyze the dimensional relations among autistic traits and the ERN moderated by each ADHD symptom cluster separately and an additional two models using the Δ ERN (see Fig. 1). In other words, these analyses were used to determine how inattention symptoms impact the relation between autistic traits and the ERN in one model and how hyperactivity-impulsivity symptoms impact this relation in the second model. For these analyses, both the autism and NTD groups were included together to obtain a broader range of autistic trait scores that more accurately represent the spectrum of autistic traits observed in the real world, which in turn, provide a more dimensional understanding of autistic traits. Analyses were repeated with the Δ ERN as the outcome. For significant moderation models, model effects were probed by testing simple-slopes at mean Low (i.e., ≤ 59) and Elevated (i.e., ≥ 60) T-scores of the corresponding model's moderator. Effects of these bootstrapped moderation models are considered statistically significant if their 95% CI does not contain 0.0 and p -values are reported for transparency.

Results

Tier 1: Preliminary Analysis and Group Differences of Primary Variables

All study variables were screened for univariate outliers, defined as values greater than $\pm 3SD$ the within-group mean. This process identified 0.14% (autism group) to 0.09% (NTD group) of data points that were corrected to the most extreme value $\pm 3SD$ within-group mean. AQ rating scale data have been reported for subsets of the current sample to examine conceptually unrelated hypotheses (Matsuba et

al., 2022). Data for study's primary outcomes (ERN, Δ ERN) and moderators (BASC-2/3 subscales) have not been previously reported. Sample demographics are shown in Table 1. No group differences across age, sex, race, or perceptual reasoning abilities (PRI) were observed (p 's > .05). The zero-order correlation matrix is shown in Table 2.

Group Differences across Rating Scale and Behavioral Measures

Inspection of Table 1 indicated that the autism group exhibited significantly higher autistic traits ($d=4.07$) compared to the NTD group. The autism group demonstrated higher and large-magnitude parent-reported inattentive ($d=2.25$) and hyperactive behaviors ($d=1.91$) compared to the NTD group. Regarding behavioral measures, the NTD group had higher accuracy scores compared to the autism group (95.6% vs 90.1%, respectively; $d=0.78$) on the Flanker task, but no RT group differences ($p > .05$).

Group Differences Across Neuro-Based Measures

Figure 2 shows the grand averages of the ERN, CRN, and Δ ERN for both the autism and NTD groups at Fz. Both the autism and NTD groups showed the expected larger and more negative activity within 0–100 ms after making an incorrect response compared to making a correct response during the arrow Flanker task. No between group differences were observed for the ERN, CRN, nor the Δ ERN mean amplitudes (p 's > .05).

Tier 2: Moderation Models

Results of the bias-corrected, bootstrapped moderation models are summarized here. A total of four separate simple moderation models were conducted to examine the moderating effects of each ADHD symptom cluster on the relation between autistic traits for both the ERN and the Δ ERN during a Flanker task. Reporting is truncated here for readability and organized by outcome; full model outputs are shown in Table 3.

Models Predicting ERN Mean Amplitude From 0–100 ms

The first model examined the potential interaction between autistic traits and inattention in predicting the ERN mean amplitude. There was a significant interaction between autistic traits and inattention ($\beta=0.01$). Among children with elevated endorsement of inattention (i.e., T-scores ≥ 60), higher autistic traits were related to attenuated ERN mean amplitudes (95%CI: 0.02–0.61; $p < .01$ Fig. 3), while for children with Low endorsed inattention (i.e., T-scores ≤ 59) there was no relation between autistic traits and the ERN (95%CI contains 0.0; $p > .05$). In the second model, the interaction between autistic traits and hyperactivity-impulsivity predicting the ERN was tested. Similarly, there was a significant interaction between autistic traits and hyperactivity-impulsivity ($\beta=0.01$). However, among both children with either Elevated or Low endorsement of hyperactive-impulsive behaviors, there was no relation between autistic traits and the ERN (95%CI's contain 0.0; p 's > .05; Fig. 3).

Table 2 Correlations among Study Variables

Variable	1	2	3	4	5	6	7	8	9	10
1. Age	–									
2. PRI	–.08	–								
3. AQ Total	.13	.20	–							
4. BASC-2/3 Attention Problems	.05	.12	.79***	–						
5. BASC-2/3 Hyperactivity	–.03	.16	.78***	.78***	–					
6. Flanker RT	–.50***	.22	–.06	.02	.08	–				
7. Flanker Acc	.17	.05	–.39*	–.26	–.30	.01	–			
8. ERN mAmp at 0–100 ms	–.40*	–.03	.02	–.03	.18	.11	–.11	–		
9. CRN mAmp at 0–100 ms	–.34*	.10	–.05	.09	–.01	–.02	.01	.17	–	
10. Δ ERN mAmp at 0–100 ms	–.15	–.10	.06	–.10	.18	.12	–.12	.76***	–.38*	–

PRI = WASI-Perceptual Reasoning Index; RT = reaction time; Acc = percent accuracy; mAmp = mean amplitude

* = $p < .05$; ** = $p < .01$; *** = $p < .001$

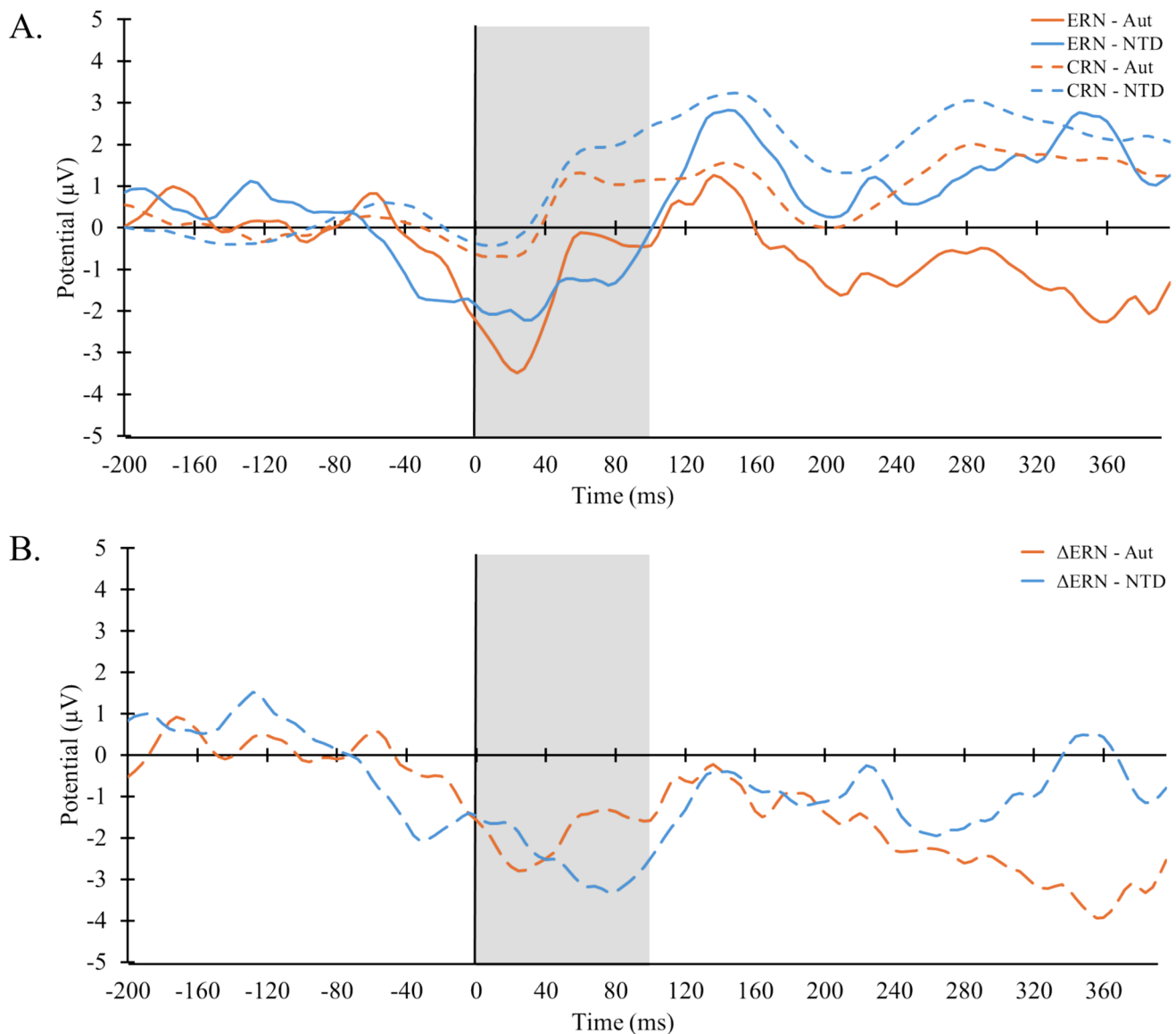


Fig. 2 Response-locked ERN, CRN, and Δ ERN waveforms at Fz. (A) Waveforms depict ERN and CRN for autism (Aut) and Neurotypically Developing (NTD) groups. (B) Δ ERN waveforms (derived from

activity during error trials minus activity during correct trials) for both autism and NTD groups. Grey shaded region across both represented ERN mean amplitude extraction of 0–100 ms time window

Models Predicting Δ ERN Mean Amplitudes From 0–100 ms

To better understand these moderating effects of ADHD symptom clusters on error-monitoring, the same two models were tested again instead using the Δ ERN mean amplitude extracted at 0–100 ms as the outcome. In the first model testing the moderating effects of inattention on the relation between autistic traits and the Δ ERN, there was a significant but small interaction between autistic traits and inattention ($\beta=0.01$). Among children with Elevated endorsement of inattention, more autistic traits were related to an attenuated ERN (95%CI: 0.02–0.61; $p<.01$; Fig. 3), while no relation

between autistic traits and the ERN was observed among children with Low endorsed inattention (95%CI contains 0.0; $p>.05$). In the second model testing the moderating effects of hyperactive-impulsive behaviors, the interaction between autistic traits and hyperactivity-impulsivity was non-significant (95%CI contains 0.0; $p=.08$). There were no significant main effects of autistic traits and hyperactivity-impulsivity (all 95%CI contain 0.0; all p 's $>.05$) within this model.

Taken together, the findings indicate that the autism group experiences more attentional and hyperactive symptoms and had moderately lower accuracy scores on the Flanker compared to NTD same-aged and IQ-matched

Table 3 Moderating Effects of Inattention and Hyperactivity on Autistic Traits and Error Monitoring (ERN measured at 0–100 ms)

Model/Predictors	Outcome: ERN mean amplitude at 0–100 ms					Outcome: Δ ERN mean amplitude at 0–100 ms				
	β	SE	p	95% CI	R ²	ΔR^2	β	SE	p	95% CI
Moderator Model: Inattention										
Intercept	-2.76	1.00	<.001	-5.172, -1.123	0.12	0.11	-3.43	0.96	<.001	-5.692, -1.847
Autistic Traits (centered)	0.08	0.08	0.27	-0.059, 0.243			0.15	0.08	0.03	0.016, 0.320
Inattention (centered)	-0.09	0.08	0.23	-0.245, 0.049			-0.16	0.08	0.05	-0.316, 0.000
Autistic Traits * Inattention	0.01	0.01	0.03	0.001, 0.024			0.01	0.01	0.03	0.001, 0.022
Moderator Model: Hyperactivity										
Intercept	-2.61	0.74	<.001	-4.215, -1.283	.05	0.06	-3.23	0.93	<.001	-5.227, -1.543
Autistic Traits (centered)	-0.07	0.07	0.41	-0.176, 0.095			-0.04	0.07	0.59	-0.171, 0.115
Hyperactivity (centered)	0.04	0.07	0.57	-0.106, 0.159			0.04	0.07	0.61	-0.112, 0.165
Autistic Traits * Hyperactivity	0.01	0.00	0.02	0.001, 0.016			0.01	0.00	0.08	0.000, 0.016

peers. Additionally, both children with autism and their NTD peers had similar RT on the Flanker task and did not show significant ERN and Δ ERN mean amplitude differences. Interestingly, inattention significantly moderated the AQ predicting both the ERN and the Δ ERN such that only children with higher endorsed inattention showed attenuated ERN and Δ ERN mean amplitudes as the number of endorsed autistic traits increased. In contrast, hyperactivity-impulsivity significantly moderated the AQ predicting the ERN model; however, when probing the interaction, neither children with higher or lower endorsed hyperactivity-impulsivity impacted the relation between the AQ and the ERN. Hyperactivity-impulsivity did not moderate the Δ ERN model.

Discussion

The present study was the first to examine the simple moderating effects of inattention and hyperactivity-impulsivity symptoms relevant to ADHD on the relation between autistic traits and error monitoring in children with and without autism. This study included a clinically evaluated and age- and IQ-matched sample and multi-method design to investigate error monitoring among children. Overall, results were consistent with prior evidence of higher rates of ADHD symptomatology among children diagnosed with autism compared to their NTD peers. This study expanded upon past literature by finding between group differences on performance-based measures and reports novel moderations of attentional problems on neural measures associated with error-monitoring. This study took a dimensional approach to understanding the interplay between autism and ADHD by examining the moderating effects of ADHD symptom clusters on the relations between autistic traits and error-monitoring in a group of children with and without autism. Primary findings suggest that inattention, but not hyperactivity-impulsivity had a moderating effect providing initial evidence that ADHD symptom clusters influence the ERN and Δ ERN. This finding is consistent with Braver's original (2012) and ongoing work linking the dual mechanism of control framework to executive functioning, which posits that cognitive control largely demands on two temporal modes: proactive (i.e., anticipating and sustaining goal-related information for optimal cognitive functioning) and reactive (i.e., a more transient and stimulus-driven process for evaluating goal-related performance). Within this context, it is likely that the Flanker task elicits higher conflict monitoring demands that are more consistent with the reactive mode rather than response inhibition (proactive mode). An alternative explanation may relate to the range of scores included in the study for inattention

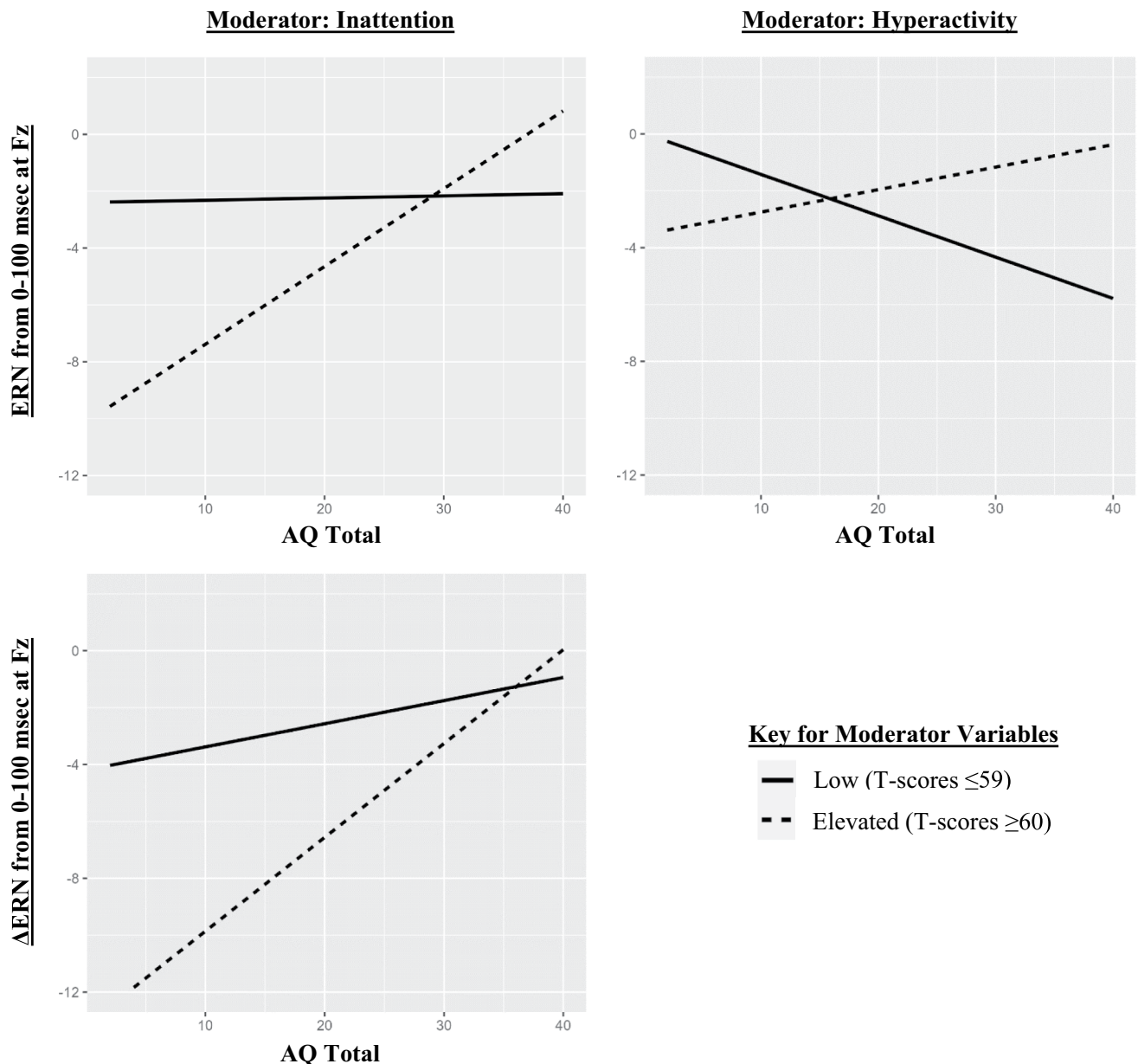


Fig. 3 Simple Slopes of Significant Interaction Effects. For both models with Inattention moderator models, the significant positive association between AQ total and the ERN becomes evident when Inatten-

tion is Elevated. For the Hyperactivity moderator model, the relation between AQ total and ERN is non-significant when Hyperactivity is either Low or Elevated

and hyperactivity problems. Autism and inattentive symptoms tend to be more closely coupled in autistic people with lower support needs, while hyperactivity concerns are more likely to be observed in individuals with higher support needs. Overall, the finding that inattention moderates the relation between autistic traits and the ERN and Δ ERN has both empirical and clinical implications for our understanding of the impacts of autistic heterogeneity on higher order cognitive processes and may help us to better understand the inconsistent ERN findings which have been reported in the autism literature. Among children with

high autistic traits, elevated inattention was found to blunt both ERN and Δ ERN mean amplitudes. Further, ERN and Δ ERN mean amplitudes were not impacted by increasing autistic traits among children with low inattention. The finding that the ERN and Δ ERN were not impacted by autistic traits among those with few inattention symptoms is in line with the current literature that suggests that the ERN and the Δ ERN generally do not differ between autistic and non-autistic children. These results are consistent with the literature focused on ADHD, which finds that both a diagnosis of ADHD (Geburek et al., 2013) and increased ADHD traits

(including inattention), (Czobor, et al., 2016; Marquardt et al., 2018) are each generally associated with a smaller ERN. While the data in Fig. 3 suggests that high inattention and low autistic traits are associated with an increased ERN, this finding is largely driven by participants having either high autistic and high inattention scores or low autistic traits and low inattention scores. These findings parallel the work of others who note that higher anxiety scores lead to higher ERN amplitudes within groups of autistic and non-autistic children (Rosen & Lerner, 2018). Together these findings provide preliminary support for the notion that error-monitoring is not a fundamental processing difference between autistic and non-autistic youth, but rather that it is impacted by relevant individual difference factors such as ADHD symptom severity. This idea is particularly important as it suggests that the use of the ERN as a potential biomarker for early identification of autism could be refined by accounting for underlying ADHD symptoms.

Hyperactivity-impulsivity was not a significant moderator of the relation between the ERN and autistic traits. Specifically, hyperactivity-impulsivity only moderated the relation between autistic traits and the ERN, though probing the simple slopes of these interactions at mean Low and Elevated hyperactivity scores yielded no significant results for these simple-slope points tested. No studies of autism have examined ADHD symptoms categorically nor dimensionally as they relate to the ERN and as such these data await replication. Only one study of children with ADHD to date has examined the relation between the ERN and defined ADHD presentations. Liu et al., 2020, found no differences in the ERN across those with ADHD predominantly inattentive, predominantly hyperactive and combined presentations. When examined dimensionally, Czobor et al., 2016 found that both inattention and hyperactivity-impulsivity as measured by the ADHD Symptoms Rating Scale (ASRS) were negatively correlated with ERN amplitude in a group of adults with and without ADHD. This discrepancy suggests that dimensional approaches might be particularly beneficial to understanding how different facets of symptoms relevant to ADHD map on to neural mechanisms. Of note, Liu focused on children with Czobor focused on adults, and age has been shown to impact the amplitude of the ERN among NTD children (Boen et al., 2022) as well as among those with ADHD (Liu et al., 2020) suggesting development is an important factor in deconstructing the impacts of dimensionality on individual differences. In addition to the novel moderation effects presented here, the data are informative with respect to our categorical understanding of error monitoring in autism.

Regarding Flanker task group differences, the autistic group responded as quickly but less accurately compared to same-aged and IQ-matched NTD peers. Extant literature

examining performance on Flanker measures among children with autism has largely been mixed, with some studies demonstrating lower accuracy and/or RT compared to neurotypically developing peers (Mutreja et al., 2015; Pudło & Pisula, 2022) while others show no between group differences (Boland et al., 2018; Henderson, 2006). Possible explanations for these mixed findings include differences in sample group characterization requirements (i.e., formal diagnoses, cut-off scores, self-reported prior diagnoses), age ranges (i.e., broad/narrow age ranges), task-parameters (i.e., stimuli type/durations), and controlling for confounding variables (i.e., IQ-matching). The current study contributes to the literature by replicating prior evidence that children with autism do not show RT differences on a traditional arrow flanker task (Henderson et al., 2006; South et al., 2010) and demonstrate lower accuracy scores (Adams & Jarrold, 2011; Faja et al., 2016) using a discrete, phenotypic sample of children between the ages of 10–17 years old, formally diagnosed with autism, and their age- and IQ-matched peers.

Regarding neural measure, there were no group differences in the amplitude of the ERN nor the Δ ERN for the 0–100 ms window. These findings are consistent with studies of the ERN among autistic children in the same general age as the current study (Groen, 2008; Henderson et al., 2015) that focused on visual processing. However, there is significant variability in the field with respect to critical EEG data extraction parameters that impact the interpretation of ERP research findings. Specifically, ERN time window differs significantly across studies with some starting their time windows just before or at responses onset, and others starting between 40 and 60 ms post responses. These methodological decisions can have large impacts on the presence and direction of group differences as can be seen by visually examining the onset of waveform negativities in Fig. 2b. Future studies should aim to focus on extraction windows beginning earlier (Luck et al., 2018) and mean amplitudes in characterizing ERPs in autism.

Limitations and Future Directions

This study provided important initial evidence for the moderating effects of ADHD symptoms on errors monitoring, and there are key next steps for the continuation of this research. The present study's sample size, though adequately powered, was relatively small, homogenous in its diversity, and had a low threshold of four error trials for acceptance. As such, future work should seek to replicate these findings in a larger, more diverse sample with more error trials. Given that children with autism experience elevated rates of inattention and hyperactivity-impulsivity (Yerys et al., 2009),

which then moderate the relation between autistic traits and error monitoring, future work is needed to determine the extent to which ADHD symptoms are implicated in other related behavioral outcomes. Research comparing children with autism, children with ADHD, children with co-occurring ADHD and autism, and NTD children would serve to further elucidate how ADHD symptoms affect error monitoring among children. Autism and ADHD are highly heterogeneous conditions that commonly co-occur with other symptoms and conditions, including anxiety that should be considered in future work.

While the two groups were age and IQ matched; Table 2 showed significant associations among the Δ ERN and the flanker RTs. The effects of age could not be controlled for in the current study due to power concerns. Given notable developmental differences among children and adolescents, future studies should consider examining the impact of age (i.e., child vs. adolescent) among these moderating effects in a large sample.

Additionally, approximately 65% of children in the autism group reported taking medication in the current study. While the majority of mental health medications have not been associated with impacting the ERN, the effects of certain medication groups warrant consideration. For example, stimulant medication has been linked with no ERN effects (Groen et al., 2008). While less than 18% of children in the autism group were on a stimulant medication in the current study; additional studies are warranted to determine the extent to which these medications and others may effect the ERN.

Given the interacting effects of ADHD symptom clusters within children with autism, future work investigating causal underlying neurocognitive processes, such as executive functioning (Kofler et al., 2024), that contribute to the manifestation of these symptoms across both disorders is necessary. As interest around understanding shared symptomatology across autism and ADHD increases, studies should continue to investigate the utility of the Δ ERN for understanding neural differences across neurodevelopmental disorders and their co-occurrence. While the current study provides an important and novel first step towards these efforts, this line of investigation may help improve identification; advance predictive predicative and functional outcomes, and the tailoring of interventions for those with autism with and without significant attentional difficulties (Antshel & Russo, 2018).

Attentional differences need to be considered in our understanding of the neurophysiology and neural mechanisms relevant to autism. We assessed whether relevant symptom clusters impacted behavioral and neural measures of error-monitoring. Inattention significantly moderated the relationship between autistic traits and error-monitoring,

suggesting that children with autism and high inattention may have a dampened neural response to errors compared to children with low inattention. Importantly, understanding the interplay among ADHD symptoms, autistic traits, and neural markers (such as the ERN) may further refine diagnostic identification, improve prediction outcomes, and inform targeted intervention efforts across these two commonly co-occurring conditions. Moving forward, ADHD symptomologies, and inattention in particular, should be considered when formulating conclusions regarding the ERN in children with autism in addition to age and IQ.

Acknowledgements We thank Alyssa Prawl, Megan Blanc, and other previous C.A.R.E. Lab members for their contributions to data collection. We thank all the children and their families that participated in this study. We thank the Society for the Study of School Psychology (SSSP) for funding part of this research (N.S. and N.R.). This work was also financially supported by the National Institute of Mental Health (#101536) awarded to N.R. Lastly, E.F.S. was funded at the time of this study on the Postdoctoral Scholar Program at Syracuse University (# PD-2741-2021) awarded to N.R. and K.A.

Author Contributions All authors contributed to the study conception and design. Material preparation and data collection were performed by NS and NR. Data processing were performed by EFS, MS, and YK. Data analyses were completed by EFS, JN, and NR. The first draft of the manuscript was written by EFS and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Funding National Institute of Mental Health, 101536, Natalie Russo, Postdoctoral Scholar Program at Syracuse University, PD-2741—2021, Natalie Russo, Postdoctoral Scholar Program at Syracuse University, PD-2741—2021, Kevin Antshel, Society for the Study of School Psychology.

Declarations

Conflict of interest The authors have no additional conflicts of interest to report.

Ethical Approval Syracuse University Institutional Review Board (IRB) approval was obtained/maintained, and all participating caregivers and children gave informed consent/assent.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

- Adams, N. C., & Jarrold, C. (2012). Inhibition in autism: Children with autism have difficulty inhibiting irrelevant distractors but not prepotent responses. *Journal of Autism and Developmental Disorders*, 42, 1052–1063. <https://doi.org/10.1007/s10803-011-1345-3>
- Antshel, K. M., & Russo, N. (2019). Autism spectrum disorders and ADHD: Overlapping phenomenology, diagnostic issues, and treatment considerations. *Current Psychiatry Reports*, 21, 1–11. <https://doi.org/10.1007/s11920-019-1020-5>
- Auyeung, B., Baron-Cohen, S., Wheelwright, S., & Allison, C. (2008). The Autism Spectrum Quotient: Children's Version (AQ-Child). *Journal of Autism and Developmental Disorders*, 38(7), 1230–1240. <https://doi.org/10.1007/s10803-007-0504-z>
- Barkley, R. A. (2011). The important role of executive functioning and self-regulation in ADHD. *J Child Neuropsychol*, 113(21), 41–56.
- Baron-Cohen, S., Hockstra, R. A., Knickmeyer, R., & Wheelwright, S. (2006). The Autism-Spectrum Quotient (AQ)—Adolescent version. *Journal of Autism and Developmental Disorders*, 36(3), 343–350. <https://doi.org/10.1007/s10803-006-0073-6>
- Baron-Cohen, S., Wheelwright, S., Skinner, R., Martin, J., & Clubley, E. (2001). The autism-spectrum quotient (AQ): Evidence from Asperger syndrome/high-functioning autism, males and females, scientists and mathematicians. *Journal of Autism and Developmental Disorders*, 31(1), 5–17. <https://doi.org/10.1023/a:1005653411471>
- Boen, R., Quintana, D. S., Ladouceur, C. D., & Tamnes, C. K. (2022). Age-related differences in the error-related negativity and error positivity in children and adolescents are moderated by sample and methodological characteristics: A meta-analysis. *Psychophysiology*, 59(6), Article e14003. <https://doi.org/10.1111/psyp.14003>
- Bogte, H., Flamma, B., Van Der Meere, J., & Van Engeland, H. (2007). Post-error adaptation in adults with high functioning autism. *Neuropsychologia*, 45(8), 1707–1714. <https://doi.org/10.1016/j.neuropsychologia.2006.12.020>
- Boland, K. M., Stichter, J. P., Beversdorf, D. Q., & Christ, S. E. (2019). Brief report: Flanker visual filtering ability in older adolescents with autism spectrum disorder. *Journal of Autism and Developmental Disorders*, 49, 422–428. <https://doi.org/10.1007/s10803-018-3755-y>
- Botvinick, M. M., Cohen, J. D., & Carter, C. S. (2004). Conflict monitoring and anterior cingulate cortex: An update. *Trends in Cognitive Sciences*, 8(12), 539–546. <https://doi.org/10.1016/j.tics.2004.10.003>
- Brainard, D. H. (1997). The Psychophysics Toolbox. *Spatial Vision*, 10(4), 433–436. <https://doi.org/10.1163/156856897X00357>
- Braver, T. S. (2012). The variable nature of cognitive control: A dual mechanisms framework. *Trends in Cognitive Sciences*, 16(2), 106–113.
- Braver, T. S., Kizhner, A., Tang, R., Freund, M. C., & Etzel, J. A. (2021). The dual mechanisms of cognitive control project. *Journal of Cognitive Neuroscience*, 33(9), 1990–2015.
- Burack, J. A., Iarocci, G., Flanagan, T. D., & Bowler, D. M. (2004). On mosaics and melting pots: Conceptual considerations of comparison and matching strategies. *Journal of Autism and Developmental Disorders*, 34(1), 65–73. <https://doi.org/10.1023/B:JADD.0000018076.90715.00>
- Burgio-Murphy, A., Klorman, R., Shaywitz, S. E., Fletcher, J. M., Marchione, K. E., Holahan, J., Stuebing, K. K., Thatcher, J. E., & Shaywitz, B. A. (2007). Error-related event-related potentials in children with attention-deficit hyperactivity disorder, oppositional defiant disorder, reading disorder, and math disorder. *Biological Psychology*, 75(1), 75–86. <https://doi.org/10.1016/j.biopsycho.2006.12.003>
- Cavanagh, J. F., & Shackman, A. J. (2015). Frontal midline theta reflects anxiety and cognitive control: Meta-analytic evidence. *Journal of Physiology-Paris*, 109(1–3), 3–15. <https://doi.org/10.1016/j.jphysparis.2014.04.003>
- Centers for Disease Control and Prevention. (2023). Data and statistics on children's mental health. Retrieved January, 30, 2023.
- Czobor, P., Kakuszi, B., Németh, K., Balogh, L., Papp, S., Tombor, L., & Bitter, I. (2017). Electrophysiological indices of aberrant error-processing in adults with ADHD: A new region of interest. *Brain Imaging and Behavior*, 11, 1616–1628.
- Davies, P. L., Segalowitz, S. J., & Gavin, W. J. (2004). Development of response-monitoring ERPs in 7- to 25-year-olds. *Developmental Neuropsychology*, 25(3), 355–376. https://doi.org/10.1207/s15326942dn2503_6
- Delorme, A., & Makeig, S. (2004). EEGLAB: An open source toolbox for analysis of single-trial EEG dynamics including independent component analysis. *Journal of Neuroscience Methods*, 134(1), 9–21. <https://doi.org/10.1016/j.jneumeth.2003.10.009>
- American Psychiatric Association. (2013). Neurodevelopmental Disorders. In *Diagnostic and statistical manual of mental disorders* (5th ed.). <https://doi.org/10.1176/appi.books.9780890425596>
- American Psychiatric Association. (2022). Neurodevelopmental Disorders. In *Diagnostic and statistical manual of mental disorders* (5th ed., text rev.). <https://doi.org/10.1176/appi.books.9780890425787>
- Downes, M., Bathelt, J., & De Haan, M. (2017). Event-related potential measures of executive functioning from preschool to adolescence. *Developmental Medicine & Child Neurology*, 59(6), 581–590. <https://doi.org/10.1111/dmcn.13395>
- Faja, S., Clarkson, T., & Webb, S. J. (2016). Neural and behavioral suppression of interfering flankers by children with and without autism spectrum disorder. *Neuropsychologia*, 93, 251–261. <https://doi.org/10.1016/j.neuropsychologia.2016.10.017>
- Falkenstein, M., Hohnsbein, J., Hoormann, J., & Blanke, L. (1991). Effects of crossmodal divided attention on late ERP components. II. error processing in choice reaction tasks. *Electroencephalography and Clinical Neurophysiology*, 78(6), 447–455. [https://doi.org/10.1016/0013-4694\(91\)90062-9](https://doi.org/10.1016/0013-4694(91)90062-9)
- Geburek, A. J., Rist, F., Gediga, G., Stroux, D., & Pedersen, A. (2013). Electrophysiological indices of error monitoring in juvenile and adult attention deficit hyperactivity disorder (ADHD)—A meta-analytic appraisal. *International Journal of Psychophysiology*, 87(3), 349–362. <https://doi.org/10.1016/j.ijpsycho.2012.08.006>
- Gehring, W. J., Goss, B., Coles, M. G. H., Meyer, D. E., & Donchin, E. (1993). A neural system for error detection and compensation. *Psychological Science*, 4(6), 385–390. <https://doi.org/10.1111/j.1467-9280.1993.tb00586.x>
- Gehring, W. J., Liu, Y., Orr, J. M., & Carp, J. (2012). The error-related negativity (ERN/Ne). In S. J. Luck & E. S. Kappenman (Eds.), *The Oxford handbook of event-related potential components* (pp. 231–291). Oxford University Press.
- Gomez, R., & Sanson, A. V. (1994). Effects of experimenter and mother presence on the attentional performance and activity of hyperactive boys. *Journal of Abnormal Child Psychology*, 22(5), 517–529. <https://doi.org/10.1007/BF02168935>
- Groen, Y., Wijers, A. A., Mulder, L. J. M., Waggeveld, B., Minderaa, R. B., & Althaus, M. (2008). Error and feedback processing in children with ADHD and children with autistic spectrum disorder: An EEG event-related potential study. *Clinical Neurophysiology*, 119(11), 2476–2493. <https://doi.org/10.1016/j.clinph.2008.08.004>
- Grzadzinski, R., Dick, C., Lord, C., & Bishop, S. (2016). Parent-reported and clinician-observed autism spectrum disorder (ASD) symptoms in children with attention deficit/hyperactivity disorder (ADHD): Implications for practice under DSM-5. *Molecular Autism*, 7(1), 7. <https://doi.org/10.1186/s13229-016-0072-1>

- Hajcak, G., Klawohn, J., & Meyer, A. (2019). The utility of event-related potentials in clinical psychology. *Annual Review of Clinical Psychology*, 15, 71–95. <https://doi.org/10.1146/annurev-clinpsy-050718-095457>
- Henderson, H., Schwartz, C., Mundy, P., Burnette, C., Sutton, S., Zahka, N., & Pradella, A. (2006). Response monitoring, the error-related negativity, and differences in social behavior in autism. *Brain and Cognition*, 61(1), 96–109. <https://doi.org/10.1016/j.bandc.2005.12.009>
- Herrmann, M. J., Saathoff, C., Schreppel, T. J., Ehrlis, A.-C., Scheuerpflug, P., Pauli, P., & Fallgatter, A. J. (2009). The effect of ADHD symptoms on performance monitoring in a non-clinical population. *Psychiatry Research*, 169(2), 144–148. <https://doi.org/10.1016/j.psychres.2008.06.015>
- Hill, E. L. (2004). Evaluating the theory of executive dysfunction in autism. *Developmental Review*, 24(2), 189–233. <https://doi.org/10.1016/j.dr.2004.01.001>
- Jadhav, C., Kamble, P., Mundewadi, S., Jaiswal, N., Mali, S., Ranga, S., Suvvari, T. K., & Rukadikar, A. (2022). Clinical applications of EEG as an excellent tool for event related potentials in psychiatric and neurotic disorders. *International Journal of Physiology, Pathophysiology and Pharmacology*, 14(2), 73.
- JASP Team (2024). JASP (Version 0.18.3)[Computer software].
- Jung, T. P., Makeig, S., Humphries, C., Lee, T. W., McKeown, M. J., Iragui, V., & Sejnowski, T. J. (2000). Removing electroencephalographic artifacts by blind source separation. *Psychophysiology*, 37(2), 163–178. <https://doi.org/10.1111/1469-8986.3720163>
- Kaiser, A., Aggensteiner, P.-M., Baumeister, S., Holz, N. E., Banaschewski, T., & Brandeis, D. (2020). Earlier versus later cognitive event-related potentials (ERPs) in attention-deficit/hyperactivity disorder (ADHD): A meta-analysis. *Neuroscience & Biobehavior Reviews*, 112, 117–134. <https://doi.org/10.1016/j.neubiorev.2020.01.019>
- Kappenman, E. S., Farrens, J. L., Luck, S. J., & Proudfit, G. H. (2014). Behavioral and ERP measures of attentional bias to threat in the dot-probe task: Poor reliability and lack of correlation with anxiety. *Frontiers in Psychology*, 5. <https://doi.org/10.3389/fpsyg.2014.01368>
- Kaufman, J., Birmaher, B., Brent, D., Rao, U., Flynn, C., Moreci, P., Williamson, D., & Ryan, N. (1997). Schedule for affective disorders and schizophrenia for school-age children-present and lifetime version (K-SADS-PL): Initial reliability and validity data. *Journal of the American Academy of Child & Adolescent Psychiatry*, 36(7), 980–988. <https://doi.org/10.1097/00004583-199707000-00021>
- Lopez-Calderon, J., & Luck, S. J. (2014). ERPLAB: An open-source toolbox for the analysis of event-related potentials. *Frontiers in Human Neuroscience*, 8. <https://doi.org/10.3389/fnhum.2014.00213>
- Lord, C., Rutter, M., DiLavore, P. C., Risi, S., Gotham, K., & Bishop, S. (2012). *Autism diagnostic observation schedule, second edition*. Western Psychological Services.
- LoTemplio, S. B., Lopes, C. L., McDonnell, A. S., Scott, E. E., Payne, B. R., & Strayer, D. L. (2023). Updating the relationship of the Ne/ERN to task-related behavior: A brief review and suggestions for future research. *Frontiers in Human Neuroscience*, 17, Article 1150244. <https://doi.org/10.3389/fnhum.2023.1150244>
- Luck, S. J. (2005). *An introduction to the event-related potential technique*. MIT Press.
- Marquardt, L., Eichele, H., Lundervold, A. J., Haavik, J., & Eichele, T. (2018). Event-related-potential (ERP) correlates of performance monitoring in adults with attention-deficit hyperactivity disorder (ADHD). *Frontiers in Psychology*, 9, Article 485. <https://doi.org/10.3389/fpsyg.2018.00485>
- Marsh, C. L., Groves, N. B., Mehra, L. M., Black, K. E., Irwin Harper, L. N., Meyer, A., & Kofler, M. J. (2023). The relation between executive functions, error-related brain activity, and ADHD symptoms in clinically evaluated school-aged children. *Child Neuropsychology*, 29(8), 1362–1387. <https://doi.org/10.1080/09297049.2023.2166029>
- Mathewson, K. J., Dywan, J., & Segalowitz, S. J. (2005). Brain bases of error-related ERPs as influenced by age and task. *Biological Psychology*, 70(2), 88–104. <https://doi.org/10.1016/j.biopsycho.2004.12.005>
- MATLAB. (2021). *version 9.10.0 (R2021a)*. Natick, Massachusetts: The MathWorks Inc.
- Matsuba, E. S., Prieve, B. A., Cary, E., Pacheco, D., Madrid, A., McKernan, E., Matsuba, E. S. M., Kaplan-Kahn, E., & Russo, N. (2024). A preliminary study characterizing subcortical and cortical auditory processing and their relation to autistic traits and sensory features. *Journal of Autism and Developmental Disorders*. <https://doi.org/10.1007/s10803-022-05773-y>
- McDermott, J. M., Perez-Edgar, K., Henderson, H. A., Chronis-Tuscano, A., Pine, D. S., & Fox, N. A. (2009). A history of childhood behavioral inhibition and enhanced response monitoring in adolescence are linked to clinical anxiety. *Biological Psychiatry*, 65(5), 445–448. <https://doi.org/10.1016/j.biopsycho.2008.10.043>
- Meyer, A., Bress, J. N., & Proudfit, G. H. (2014). Psychometric properties of the error-related negativity in children and adolescents. *Psychophysiology*, 51(7), 602–610. <https://doi.org/10.1111/psyp.12208>
- Meyer, A., & Hajcak, G. (2019). A review examining the relationship between individual differences in the error-related negativity and cognitive control. *International Journal of Psychophysiology*, 144, 7–13. <https://doi.org/10.1016/j.ijpsycho.2019.07.005>
- Meyer, A., Weinberg, A., Klein, D. N., & Hajcak, G. (2012). The development of the error-related negativity (ERN) and its relationship with anxiety: Evidence from 8 to 13 year-olds. *Developmental Cognitive Neuroscience*, 2(1), 152–161. <https://doi.org/10.1016/j.dcn.2011.09.005>
- Mutreja, R., Craig, C., & O'Boyle, M. W. (2016). Attentional network deficits in children with autism spectrum disorder. *Developmental Neurorehabilitation*, 19(6), 389–397. <https://doi.org/10.3109/17518423.2015.1017663>
- Perkins, E. R., Joyner, K. J., Foell, J., Drislane, L. E., Brislin, S. J., Frick, P. J., Yancey, J. R., Soto, E. F., Ganley, C. M., Keel, P. K., Sica, C., Flor, H., Nees, F., Banaschewski, T., Bokde, A. L. W., Desrivieres, S., Grigis, A., Garavan, H., Gowland, P., Heinz, A., ... Patrick, C. J. (2022). Assessing general versus specific liability for externalizing problems in adolescence: Concurrent and prospective prediction of symptoms of conduct disorder, ADHD, and substance use. *Journal of psychopathology and clinical science*, 131(7), 793–807. <https://doi.org/10.1037/abn0000743>
- Perrin, F., Pernier, J., Bertrand, O., Giard, M. H., & Echallier, J. F. (1987). Mapping of scalp potentials by surface spline interpolation. *Electroencephalography and Clinical Neurophysiology*, 66(1), 75–81. [https://doi.org/10.1016/0013-4694\(87\)90141-6](https://doi.org/10.1016/0013-4694(87)90141-6)
- Potts, G. F., George, M. R. M., Martin, L. E., & Barratt, E. S. (2006). Reduced punishment sensitivity in neural systems of behavior monitoring in impulsive individuals. *Neuroscience Letters*, 397(1–2), 130–134. <https://doi.org/10.1016/j.neulet.2005.12.003>
- Pudlo, M., & Pisula, E. (2022). Efficiency of attentional processes in attention network theory and autistic symptoms in adolescents with autism spectrum disorder. *Frontiers in Psychiatry*, 13, Article 950245. <https://doi.org/10.3389/fpsyg.2022.950245>
- Reynolds, C. R., & Kamphaus, R. W. (2004). *Behavior assessment system for children* (2nd ed.). Pearson Assessments.
- Reynolds, C. R., & Kamphaus, R. W. (2015). *Behavior Assessment System for Children—Third Edition, Student Observation System*. Pearson.
- Riesel, A., Weinberg, A., Endrass, T., Meyer, A., & Hajcak, G. (2013). The ERN is the ERN is the ERN? Convergent

- validity of error-related brain activity across different tasks. *Biological Psychology*, 93(3), 377–385. <https://doi.org/10.1016/j.biopsycho.2013.04.007>
- Rong, Y., Yang, C.-J., Jin, Y., & Wang, Y. (2021). Prevalence of attention-deficit/hyperactivity disorder in individuals with autism spectrum disorder: A meta-analysis. *Research in Autism Spectrum Disorders*, 83, Article 101759. <https://doi.org/10.1016/j.rasd.2021.101759>
- Rosen, T. E., & Lerner, M. D. (2018). Error-related brain activity and anxiety symptoms in youth with autism spectrum disorder. *Autism Research*, 11(2), 342–354.
- Russell, J. (Ed.). (1997). *Autism as an executive disorder*. Oxford University Press.
- Rutter, M., Le Couteur, A., & Lord, C. (2003). *Autism Diagnostic Interview-Revised*. Western Psychological Services.
- Shiels, K., & Hawk, L. W. (2010). Self-regulation in ADHD: The role of error processing. *Clinical Psychology Review*, 30(8), 951–961. <https://doi.org/10.1016/j.cpr.2010.06.010>
- Simons, R. F. (2010). The way of our errors: Theme and variations. *Psychophysiology*, 47(1), 1–14. <https://doi.org/10.1111/j.1469-8986.2009.00929.x>
- Sokhadze, E., Baruth, J., El-Baz, A., Horrell, T., Sokhadze, G., Carroll, T., Tasman, A., Sears, M., & Casanova, M. F. (2010). Impaired error monitoring and correction function in autism. *Journal of Neurotherapy*, 14(2), 79–95.
- South, M., Larson, M. J., Krauskopf, E., & Clawson, A. (2010). Error processing in high-functioning autism spectrum disorders. *Biological Psychology*, 85(2), 242–251. <https://doi.org/10.1016/j.biopsycho.2010.07.009>
- Sullivan, J. R., & Riccio, C. A. (2006). An empirical analysis of the BASC frontal lobe/executive control scale with a clinical sample. *Archives of Clinical Neuropsychology*, 21(5), 495–501. <https://doi.org/10.1016/j.acn.2006.05.008>
- Thakkar, K. N., Polli, F. E., Joseph, R. M., Tuch, D. S., Hadjikhani, N., Barton, J. J. S., & Manoach, D. S. (2008). Response monitoring, repetitive behaviour and anterior cingulate abnormalities in autism spectrum disorders (ASD). *Brain*, 131(9), 2464–2478. <https://doi.org/10.1093/brain/awn099>
- Wechsler, D., & Hsiao-pin, C. (2011). *WASI-II: Wechsler abbreviated scale of intelligence*. Pearson.
- Wiersema, J. R., Van Der Meere, J. J., & Roeyers, H. (2009). ERP correlates of error monitoring in adult ADHD. *Journal of Neural Transmission*, 116(3), 371–379. <https://doi.org/10.1007/s00702-008-0165-x>
- Wyble, B. (2019). Stream. Retrieved from osf.io/tdvxn
- Yerys, B. E., Wallace, G. L., Sokoloff, J. L., Shook, D. A., James, J. D., & Kenworthy, L. (2009). Attention deficit/hyperactivity disorder symptoms moderate cognition and behavior in children with autism spectrum disorders. *Autism Research*, 2(6), 322–333. <https://doi.org/10.1002/aur.103>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.